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ME3330 – Accelerometers Technical Summary

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Deliverable 1:

A string on a guitar was picked three times, and the resulting data was recorded by an accelerometer before having a fast Fourier transform (FFT) applied which converted the data into wavelengths. The output from a given test can be seen below in FIG. 1.

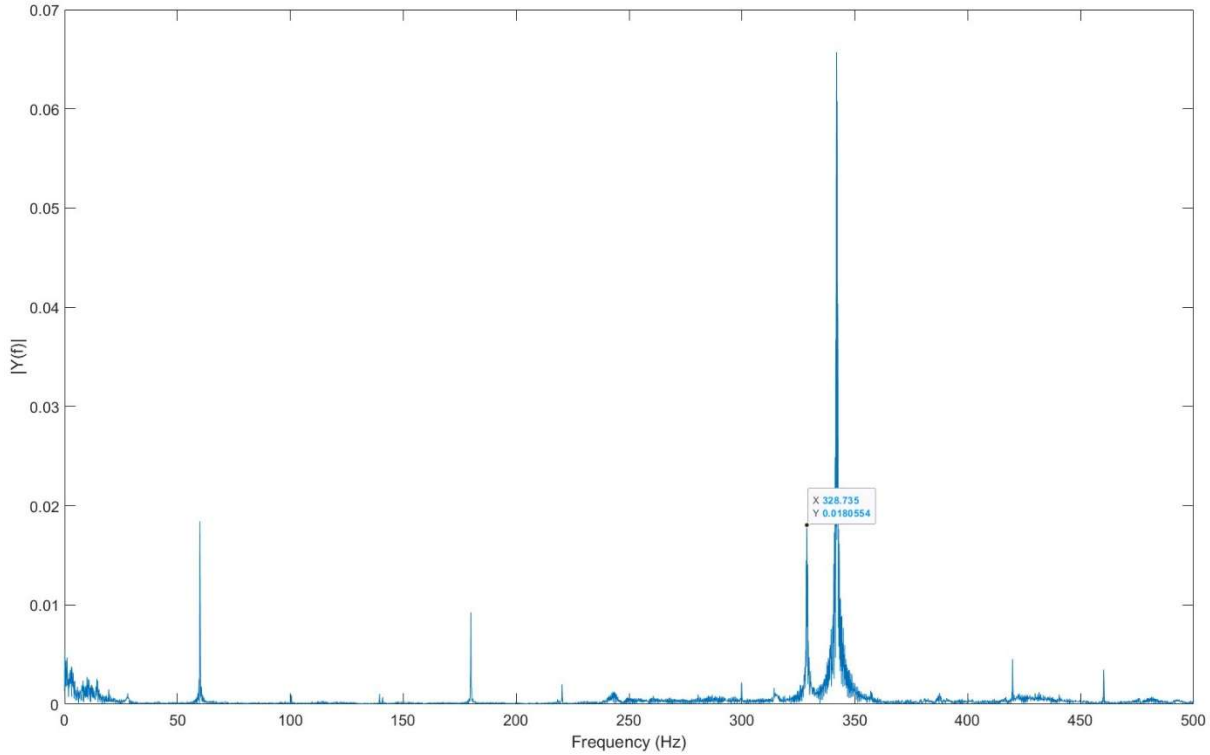


FIGURE 1: FFT of unknown guitar string. Datapoint marks frequency of interest.

These 10 tests resulted in the values plotted in TABLE 1 below.

Test Number	Frequency [Hz]
1	328.857
2	328.735
3	328.735
4	328.979
5	328.613
6	328.491
7	328.735
8	328.613
9	328.613
10	328.491

TABLE 1: Results of guitar string frequency experiment.

Using the frequencies from the 10 tests, the mean was calculated using Equation 1. This was done by adding the frequencies and dividing by the sample size.

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{1}$$

As seen below in TABLE 2, the mean frequency matches the frequency of the E string. Using the mean calculation as f_{mean}, the percent error can be calculated with Equation 2 using 329 Hz as the $f_{expected}$.

$$Percent\ Error = 100\% * \left|\frac{f_{mean} - f_{expected}}{f_{expected}}\right| \tag{2}$$

The output of the values from Equation 1 and Equation 2 are below in TABLE 2.

Mean Frequency [Hz]	Expected Frequency [Hz]	Percent Error [%]
328.674	329	0.010

TABLE 2: Calculated values from TABLE 1.

Major Talking Point 1: As seen in TABLE 1, all ten frequencies are very close together. The experiment was repeated 10 times, and therefore the extreme similarity of the data suggests that there is a strong connection between the frequency of the vibrations, and the string played on the guitar.

Major Talking Point 2: This is emphasized by the fact that in TABLE 2, the percentage error is extremely small. The expected frequency of 329 was closely matched by the experimental data. It is important to note that guitars may not be exactly tuned to 329 Hz and may output a slightly different value. The data suggests that this guitar falls slightly below 329 Hz.

Deliverable 2:

The next procedure was to study two motors vibrating inside of an acrylic shell. These were taken at three different sampling rates. Further information on sampling rates is found in Deliverable 3. FIG 3, 4, 5 are the three different sampling rates. The highest amplitude points have data points for further analysis.

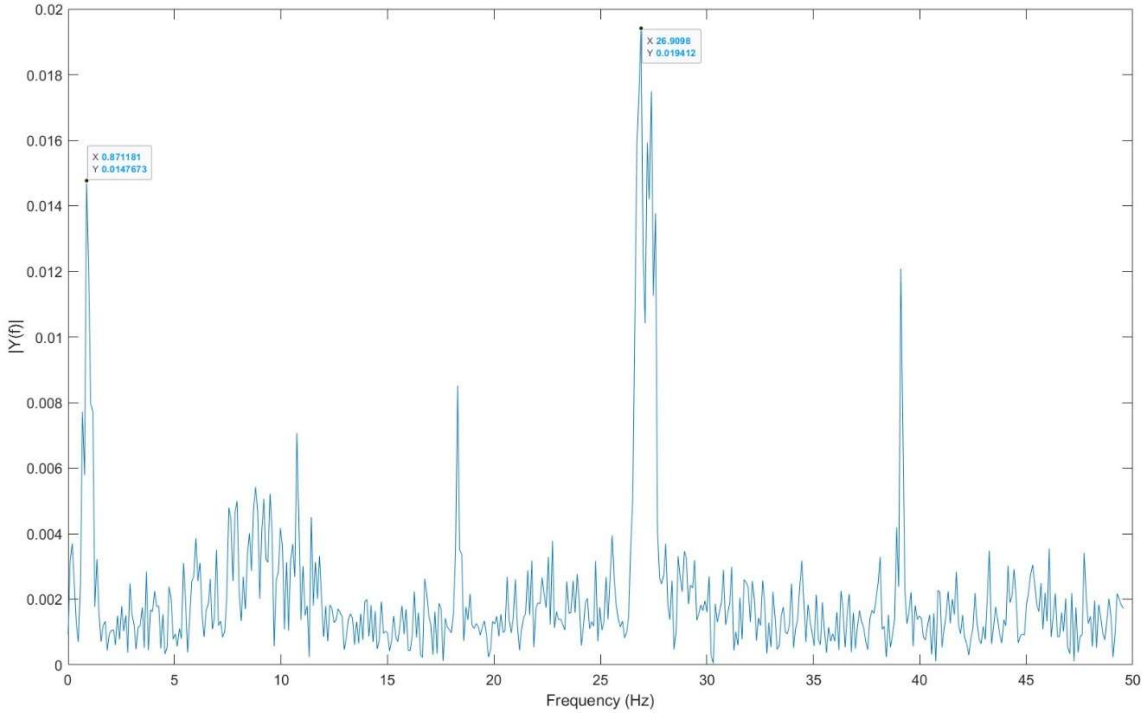


FIGURE 2: Unfiltered FFT at 0.5 fm,h sampling rate

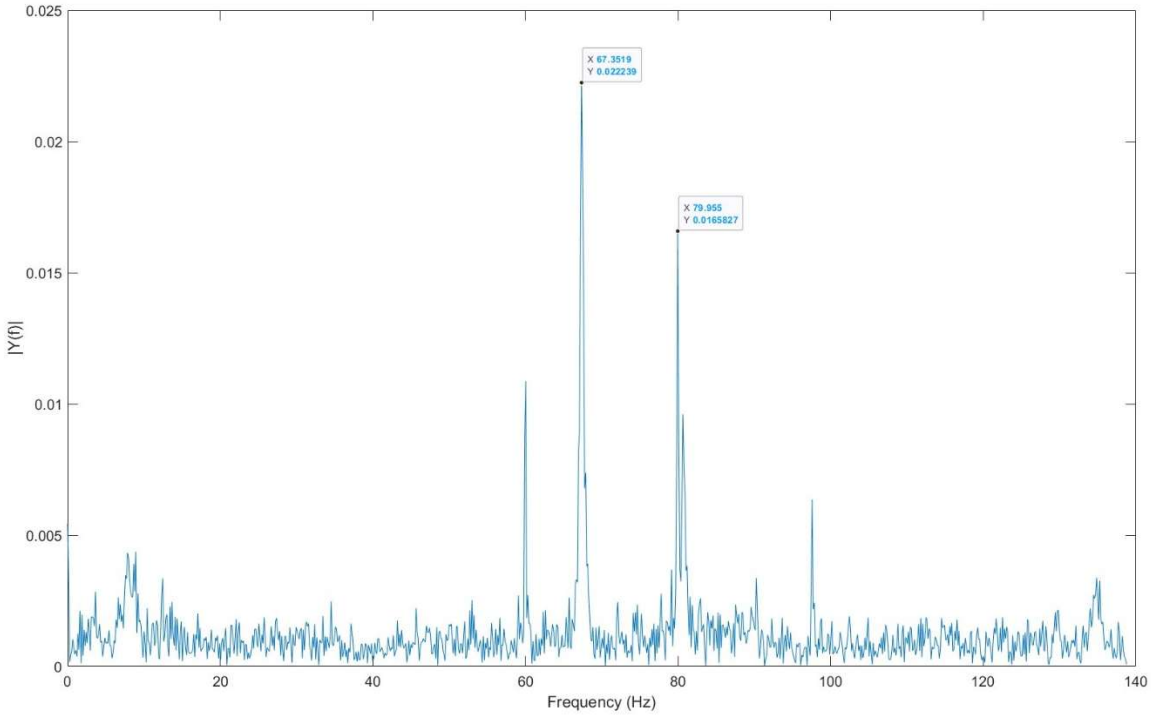


FIGURE 3: Unfiltered FFT at 1.4 fm,h sampling rate

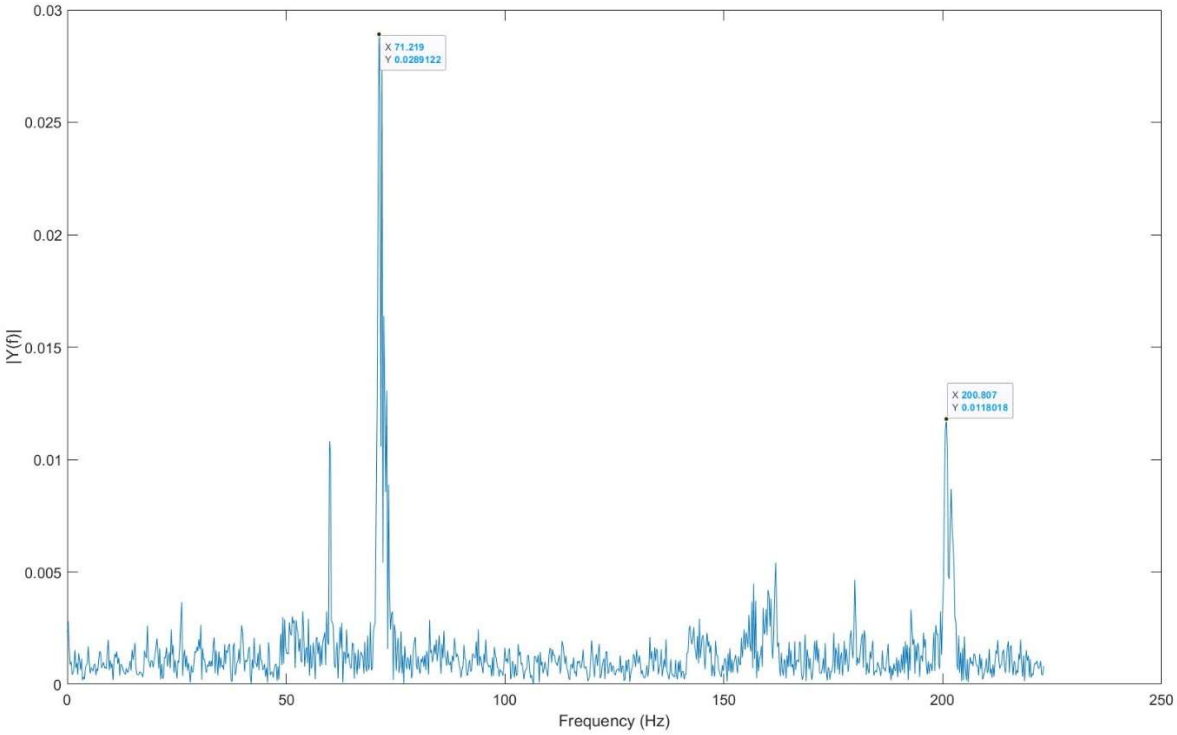


FIGURE 4: Unfiltered FFT at 2.25 fm,h sampling rate

There are two large spikes visible in FIG. 4, and these are intended to remain in place after a filter is utilized. The filtered plot of FIG. 4 is pictured below in FIG 5.

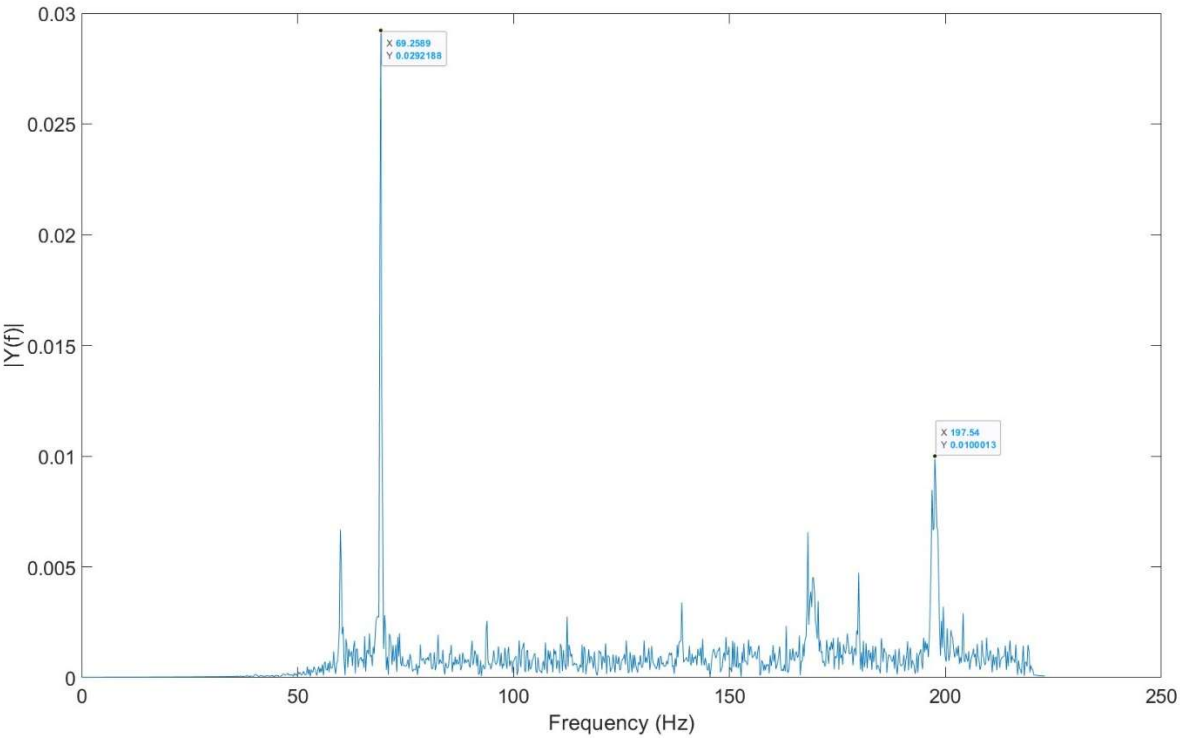


FIGURE 5: Filtered FFT using a bandpass filter

The datapoints of the sampling rate plots, as well as the filtered plot are listed below in TABLE 3.

FFT Plot	Low Frequency [Hz]	High Frequency [Hz]
Sampling Rate 0.5fm,h	0.871181	26.9098
Sampling Rate 1.4fm,h	67.3519	79.955
Sampling Rate 2.25fm,h	71.219	200.807
Filtered	69.2589	197.54

TABLE 3: High and low frequency measurements from different sampling rate experiments, as well as filtered plot.

When filtering signals, there are only a few options. A low pass or high pass filter would see the removal of either the background noise from 0-50 Hz, or 210-225Hz. This is a good option because it removes noise and leaves the peaks of both motors more isolated. The downside of using a high or low pass filter is that there must be an option to choose between trimming the front or back of our frequencies.

Thankfully, this problem can be solved with the usage of a bandpass filter. This selects a range of frequencies and only keeps the ones between the selected frequencies. This works perfectly for the frequency range seen in FIG 4, and the resulting filtered FFT seen in FIG 5 is less noisy.

Deliverable 3:

An aliased frequency occurs when a data recording is taken with too slow of a sample rate leading to loss of data. In order to calculate the frequency that this occurs for the motors, calculations will need to be performed. Beginning with calculating the sampling rates used above in FIG. 2, 3, 4, 5. These are calculating using the frequency of the high motor, $f_{m,h}$. These results can be found using Equation 3, 4, and 5.

$$\text{Sampling Rate 0.5 fm,h} = 0.5 \cdot f_{m,h} \tag{3}$$

$$\text{Sampling Rate 1.4 fm,h} = 1.4 \cdot f_{m,h} \tag{4}$$

$$\text{Sampling Rate 2.25 fm,h} = 2.25 \cdot f_{m,h} \tag{5}$$

These three sampling rates f_s are then used to solve for the Nyquist frequency (f_N) at each sampling rate. This is done using Equation 6.

$$f_N = \frac{1}{2} \cdot f_s \tag{6}$$

The Nyquist frequency is then compared to the natural frequency of the motors, f_{nat}, which were experimentally determined, and can be seen below in TABLE 4.

Motor	Natural Frequency [Hz]
Higher	198.242
Lower	72.0215

TABLE 4: Experimental results of higher and lower motor natural frequency.

The values of f_{nat} and f_N are compared with Equation 7. If the natural frequency is higher than the Nyquist frequency, then aliasing occurs.

$$f_{nat} > f_N \tag{7}$$

This procedure was performed six times, two at each of the three sampling frequencies. This resulted in three possibilities for aliasing to occur. If aliasing was predicted, then the aliasing frequency f_A would be

solved for. This is done by first solving for k, the ratio of the natural frequency to the Nyquist frequency using Equation 8.

$$k = \frac{f_{nat}}{f_N} \tag{8}$$

The final step is to use a step plot in order to solve for k'. All of the values previously solved, as well as the aliasing predictions are listed below in TABLE 5.

Scalar	f_s [Hz]	f_N [Hz]	Aliasing Higher	Aliasing Lower	k_h	k_l	k'_h	k'_l
0.5	99.121	49.5605	Yes	Yes	4	1.453	0	0.5468
1.4	277.5388	138.7694	Yes	No	1.428	N/A	0.572	N/A
2.25	446.0445	223.0223	No	No	N/A	N/A	N/A	N/A

TABLE 5: Values for calculation of aliasing frequency.

With these values, Equation 9 can be used to solve for the predicted aliasing frequency. These predictions are listed below alongside the experimental data from FIG. 2, 3, and 4 in TABLE 6.

$$f_A = k' f_N \tag{9}$$

Scalar	$f_{A,h}$ [Hz]	$f_{A,l}$ [Hz]	$f_{e,h}$ [Hz]	$f_{e,l}$ [Hz]
0.5	0	27.010	0.871181	26.9098
1.4	79.376	Not Predicted	79.955	Not Predicted
2.25	Not Predicted	Not Predicted	Not Predicted	Not Predicted

TABLE 6: Aliasing values from analysis and experimental results.

The aliasing occurs close to the predicted values. FIG. 2 shows the high motor aliasing at 0.871181 Hz, and the low motor aliasing at 26.9098 Hz, where the predicted values are 0 and 27.010 Hz respectively. This remains true for FIG 3 where the high motor aliased very close to the predicted value.

The other significant aspect is that aliasing only occurred where it was predicted to occur. There was no aliasing in FIG 3 from the lower motor, and FIG 4 had zero aliasing at all. This is important because it means that through only readings from an accelerometer, the aliasing frequency of two motors can be calculated.

Deliverable 4:

Precision and confidence intervals are both useful tools for statistics. The confidence interval is a measurement of how likely a value is to fall within a range. A confidence interval of 90% means that a value has a 90% chance of being within the given range. An increase in the confidence interval means widening the range and therefore increasing the chances of a given value falling within the range. Therefore, an increase in the confidence interval leads to a decrease in precision.

A precision interval is a measurement of the probability that a value will fall on a given interval. This can be used to measure how close a value is to its expected value; thus, it can be used to predict future values. There are clear and important distinctions between precision and confidence intervals, but both will prove useful. The precision interval will be calculated first, beginning with the standard deviation. The standard deviation is the measurement of the average distance of the datapoints from the sample mean. Standard deviation can be calculated below with Equation 10.

$$s_x = \sqrt{\frac{1}{N-1} \cdot \sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{10}$$

After standard deviation is calculated, the precision interval can be calculated with the help of the t-table at 95% for $t_{v,p}$ where v is equal to nine due to it being one less than the sample size. Finally, the mean $\bar{x}$ is taken from above.

$$x_i = \bar{x} \pm t_{v,p} \cdot s_x \tag{11}$$

The values for the precision interval are 328.6862±0.3491, which means that we can expect the next value to fall within this range. To find the confidence interval, the standard deviation of the means is needed, so Equation 12 will be used.

$$s_{\bar{x}} = \frac{s_x}{\sqrt{N}} \tag{12}$$

Next, the confidence interval is calculated using known values, and the t-table at 95%.

$$x' = \bar{x} \pm t_{v,p} \cdot s_{\bar{x}} \tag{13}$$

This yields a confidence interval of 0.1103 which means that there is a 95% chance of a value falling between the bounds of 328.5758 - 328.7965.

REFERENCES

[1] Department of Mechanical Engineering, Clemson University. *ME 3330 – Mechanical Engineering Lab II Student Manual.* Fall 2025 ed., Clemson University, 2025.

[2] mgmixing, “Acoustic Guitar Frequency Range - The Complete Guide,” *Music Guy Mixing*, May 10, 2024. https://www.musicguymixing.com/acoustic-guitar-frequency-range/